AI Image Generator

5. Module Reference

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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Stable Diffusion XL + Real-ESRGAN, served locally with Gradio, with per-image GPU telemetry recorded in SQLite.

5. Module Reference

5.1 app.py (355 lines)

Entry point of the web application.

Name	Kind	Description
MODEL_ID	constant	"stabilityai/stable-diffusion-xl-base-1.0"
OUTPUT_DIR	constant	Path to outputs/, created on import
DEFAULT_NEGATIVE	constant	"blurry, low quality, deformed, watermark, text"
HELP	constant	Markdown table of prompt words shown in the collapsible help panel
load_pipeline()	function	Loads SDXL in float16 with the fp16 variant, moves it to CUDA, enables VAE slicing, disables the console progress bar. Exits if CUDA is unavailable. Returns the pipeline.
PIPE, UPSCALER, STATIC_GPU	globals	The loaded pipeline, the Upscaler instance, and static GPU facts, created once at import.
preview(prompt)	function	Returns the Markdown "Will generate:" line for the live interpretation display.
details_markdown(rec)	function	Renders one database record as the Markdown report shown under the gallery and in History.
generate(prompt, negative, progress)	function	Runs the full pipeline for all requested images. Returns (list of PIL images, Markdown report). Raises gr.Error on an empty prompt.
history_table(search, limit)	function	Returns a pandas DataFrame with the columns shown in the History grid.
history_stats()	function	Returns the one-line aggregate summary.
refresh_history(search, limit)	function	Event handler returning (table, stats).
select_history(evt, table)	function	Row-click handler returning (image path, details Markdown). Prefers the upscaled file.

The UI is built in a gr.Blocks context with two tabs. Events wired: prompt change to preview, button click and Enter to generate, History tab select and Refresh click and search submit to refresh_history, table row select to select_history.

5.2 prompt_parser.py (252 lines)

Pure parsing, no GPU or I/O.

Name	Kind	Description
Settings	dataclass	prompt, width, height, steps, guidance, count, seed, style, upscale_to, exact_size, notes. Methods: summary() for display, upscaled_size() for the final output size.
RATIO_BUCKETS	constant	Eleven SDXL-friendly (width, height) pairs keyed by ratio string.
ORIENTATION_WORDS	constant	Ordered regex to ratio mapping; first match wins.
STYLE_SUFFIXES	constant	Ordered regex to descriptor list; first match wins; descriptors already in the prompt are not duplicated.
UPSCALE_TARGETS	constant	2k 2560, 4k 3840, uhd 3840, 8k 7680 (long side).
parse(text)	function	Returns a Settings. Order of evaluation: explicit resolution, explicit ratio, orientation word, upscale words, steps, guidance, count, seed, style, tidy-up.

Internal helpers: _take() finds a pattern in the current prompt and removes it (case-insensitive) so later patterns see the updated text; _fit_resolution() scales a requested size to about 1 megapixel and snaps to multiples of 64 within 512 to 1536; _closest_bucket() picks the nearest ratio for unusual ratios.

Running the module directly prints the interpretation of nine sample prompts.

5.3 upscaler.py (109 lines)

Name	Kind	Description
MODEL_PATH	constant	models/RealESRGAN_x4plus.pth
TILE, OVERLAP	constants	512 and 32 pixels
Upscaler(device, model_path)	class	Loads the model with spandrel, switches it to half precision on CUDA, exposes name and scale.
Upscaler.upscale_once(img)	method	One x4 pass. Iterates tiles with context padding, runs each through the network under inference_mode, crops the padding, writes into the output array.
Upscaler.upscale_to(img, long_side)	method	Repeats passes until the long side reaches the target; before a second pass resizes the intermediate so the pass lands near the target; finally Lanczos-resamples down to exactly long_side. Returns (image, info dict with upscaler, passes, from, to, seconds).

Running the module directly upscales a file given on the command line.

5.4 gpu_info.py (97 lines)

Name	Kind	Description
snapshot()	function	Runs nvidia-smi once with a fixed query and returns a dict: temperature_c, power_w, power_limit_w, gpu_util_pct, mem_util_pct, mem_used_mb, mem_total_mb, sm_clock_mhz, mem_clock_mhz, fan_pct, driver_version, pstate. Returns {} if nvidia-smi fails.
static_info()	function	Returns gpu_name, vram_total_gb, compute_capability, multiprocessors, cuda_version, cudnn_version, torch_version, diffusers_version, python_version, os.
torch_memory()	function	Returns vram_allocated_gb, vram_reserved_gb, vram_peak_gb from the torch allocator.

5.5 db.py (177 lines)

Name	Kind	Description
DB_PATH	constant	generations.db next to the module
SCHEMA	constant	CREATE TABLE IF NOT EXISTS for generations plus an index on created_at
NEW_COLUMNS	constant	Columns added after the first release; applied by ALTER TABLE when missing
connect()	function	Opens a connection with Row factory, applies schema and migrations, returns it
insert(record)	function	Inserts known keys as columns; unknown keys are packed into extra_json. Returns the row id
fetch(limit, search)	function	Newest-first list of dicts, optionally filtered by LIKE on user_prompt or final_prompt
get(row_id)	function	One record as a dict, or None
stats()	function	Aggregates: count, total and average times, fastest and slowest, average steps per second, VRAM averages and maxima, hottest temperature, peak power, upscaled count and time, first and last timestamps

Running the module directly provides the command-line viewer described in the Workflows section.

5.6 generate.py (108 lines)

Standalone command-line generator. Arguments: prompt, --negative, --steps, --guidance, --width, --height, --seed, --count, --model, --out. Loads the pipeline per invocation, enables CPU offload on cards under 12 GB, saves PNGs with timestamp and seed, and prints peak VRAM at the end. It does not write to the database.

5.7 Supporting files

File	Purpose
requirements.txt	Python dependencies with the note on installing the CUDA PyTorch build first
run-app.bat	Double-click launcher that runs app.py inside the virtual environment
README.md	User-facing quick start and prompt reference
models/RealESRGAN_x4plus.pth	Upscaler weights
docs/	This documentation set as PDFs; docs-src/ holds the Markdown sources, images and build script